



In vivo stable-isotope kinetic study suggests intracellular assembly of lipoprotein(a)

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ABSTRACT

Objective: Lipoprotein(a) [Lp(a)] consists of apolipoprotein B-100 (apoB-100) as part of an LDL-like particle and the covalently linked glycoprotein apolipoprotein(a) [apo(a)]. Detailed mechanisms of its biosynthesis, assembly, secretion and catabolism are still poorly understood. To address the Lp(a) assembly mechanism, we studied the *in vivo* kinetics of apo(a) and apoB-100 from Lp(a) and LDL apoB-100 in nine healthy probands using stable-isotope methodology.

Methods: The level of isotope enrichment was used to calculate the fractional synthesis rate (FSR), production rate (PR) and retention time (RT) using SAAMII software and multicompartmental modeling. **Results:** We observed a similar mean PR for apo(a) (1.15 nmol/kg/d) and apoB-100 (1.31 nmol/kg/d) from Lp(a), which differed significantly from the PR for apoB-100 from LDL (32.6 nmol/kg/d). Accordingly, mean FSR and RT values for Lp(a)-apo(a) were similar to those of Lp(a)-apoB and different from those for LDL-apoB.

Conclusion: Two different kinetic apoB pools within Lp(a) and LDL suggest intracellular Lp(a) assembly from apo(a) and newly synthesized LDL.

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1. Introduction

Lipoprotein(a) [Lp(a)] consists of an LDL-like particle that is covalently bound to the glycoprotein apolipoprotein(a) [apo(a)] by disulfide linkage and derives from the liver [1]. Among individuals Lp(a) plasma concentrations vary more than 1000-fold, ranging from less than 0.1 mg/dl to more than 300 mg/dl. Depending on the used genetic approach, it has been shown that between 30% and 90% of this variation in plasma Lp(a) concentration is determined by the apo(a) gene locus, encoding proteins from <300 to >800 kDa [2,3]. Apo(a) size is negatively correlated with Lp(a) concentrations,

such that low-molecular-weight (LMW) apo(a) isoforms express on average high Lp(a) plasma concentrations, while high-molecular-weight (HMW) isoforms are usually associated with lower concentrations (reviewed in [1]). Elevated plasma concentrations of Lp(a) are associated with an increased risk for developing coronary heart disease (CHD) [4,5]. Kinetic studies using radio-labeled Lp(a) indicated that the large differences in Lp(a) concentration seen among individuals are determined by synthesis and not degradation [6,7].

Little is yet known about the metabolism and physiological roles as well as the pathogenicities of Lp(a), as recently reviewed by Dubé et al. [8]. The mechanisms that control Lp(a) secretion and assembly were investigated mostly by means of hepatocyte cell culture model systems, yielding controversial results and thus remain poorly understood (see review [9]). Assembly of apo(a) and apoB to Lp(a) is generally viewed as a two-step procedure [10,11]. In a first step, distinct domains within the apoB molecule initially associate with apo(a) in a noncovalent interaction to put the two molecules

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into close proximation. In a second step, a disulfide bond is formed between apo(a) cysteine 4057 and apoB cysteine 4326 residues [10,12]. Whether this disulfide bond is linked by spontaneous oxidation or by specific enzymatic reaction is unclear [13].

Moreover, the location of this assembly process is the subject of controversial discussion as well. Intracellular, extracellular and/or plasma membrane-associated assembly procedures have been reported in various cell systems [9]. Lp(a), like many other oligomeric protein complexes, may assemble in the endoplasmic reticulum of the hepatocyte and be secreted as a whole particle [14,15]. Alternatively, newly synthesized apo(a) could bind extracellularly to preexisting LDL or VLDL circulating in the plasma. Most authors postulate an extracellular assembly of Lp(a) based on studies conducted in various cell model systems. White et al. could not detect an intracellular apo(a)–apoB complex by adding anti-apo(a) antiserum to the culture medium of primary baboon hepatocytes, but found such complexes attached to the plasma membrane. The authors therefore concluded that, at least for that particular cell culture model, Lp(a) is primarily assembled after secretion and to some extent also on the plasma membrane [16]. Similar studies in apo(a)-transfected HepG2 cells could not demonstrate an intracellular apo(a)–apoB assembly for this human hepatocyte model and thus confirmed the results of the baboon studies [10,17,18]. Nevertheless, there is also evidence for intracellular assembly of Lp(a) in cell culture. Bonen et al. were able to detect an intracellular apo(a)–apoB complex in HepG2 cells transfected with an apo(a) minigene [19].

Kinetic *in vivo* studies in humans have unfortunately also produced controversial results. Krempler et al. injected radio-labeled VLDL in Lp(a)-positive healthy probands and found no metabolic relationship between apoB of VLDL or LDL and that of Lp(a). The authors therefore concluded that Lp(a) seems to be synthesized as a separate lipoprotein independently of other apoB-containing lipoproteins [20,21]. Two *in vivo* kinetic studies using stable-isotope-labeling techniques came to the same conclusion: Morrisett et al. and Su et al. observed similar fractional synthesis rates (FSR) for Lp(a) apo(a) and Lp(a) apoB [22,23]. Whereas these findings are compatible with an intracellular assembly of nascent apo(a) and apoB to Lp(a), two other kinetic studies concluded that Lp(a) originates from *de novo* hepatic LDL as well as from plasma LDL [24,25].

We used the stable-isotope technology to investigate the metabolism of apo(a) and apoB-100, the two major Lp(a) proteins, in comparison to that of apoB from LDL in nine healthy probands. The metabolic data accumulating in this study after appropriate modeling present a scenario of virtually complete intracellular assembly of Lp(a).

2. Materials and methods

2.1. Study subjects

Nine healthy men with Lp(a) plasma concentrations >15 mg/dl were recruited from the laboratory staff at Innsbruck Medical University. They had no evidence of any chronic illness, including endocrine, hepatic, renal, thyroid, or cardiac dysfunction, and had normal body mass indices. They did not take any medication known to affect lipid metabolism. The experimental protocol was approved by the Ethics Committee of Innsbruck Medical University. All subjects participating in this study gave written informed consent.

2.2. Laboratory parameters

Total cholesterol, HDL cholesterol, LDL cholesterol and triglycerides were determined with commercially available kits from

Roche Diagnostics GmbH (Mannheim, Germany). Lp(a) was quantified, essentially as described [26], with a double-antibody ELISA using an affinity-purified anti-apo(a) antibody for coating and the horseradish peroxidase-conjugated monoclonal antibody 1A2 for detection. Lp(a)-positive plasma from Technoclone (Vienna, Austria) served as standard. To calculate PR for apo(a) and apo B from Lp(a), concentrations of Lp(a) had to be expressed in molar units. The total molecular weight of Lp(a) was calculated by adding the molecular weight of the LDL moiety of Lp(a) [27] to the mass of the various apo(a) isoforms, which was derived from the amino acid composition of the protease domain, kringle V, kringle IV repeats and the carbohydrate moiety assumed to be 30% of the apo(a) molecule [28,29].

Apo(a) phenotyping was performed by SDS agarose gel electrophoresis of plasma under reducing conditions, followed by immunoblotting as previously described [30]. A serum pool from various donors expressing apo(a) isoforms with 14, 19, 22, 27 and 35 kringle IV repeats was used as apo(a) isoform standard.

Apolipoprotein B concentrations of plasma, VLDL and LDL were measured by ELISA [31]. An affinity-purified rabbit polyclonal antibody against apoB was used for coating and the same antibody, labeled with horseradish peroxidase, for detection. A calibrated standard (Apoproteins Human ApoB, Technoclone, Vienna, Austria) served as secondary standard. Molar concentrations of apoB-100 were calculated assuming a molecular weight of 550 kDa for apoB-100.

Apolipoprotein E phenotyping was performed on delipidated plasma essentially as described [32].

2.3. Kinetic study

The kinetics of apoB in VLDL, LDL and Lp(a) as well as of apo(a) were investigated according to a previously published protocol [33]. After a 12-h overnight fast subjects were given L-[5,5,5-²H₃] leucine (d₃-leucine, 95% pure, Cambridge Isotopes Laboratories Inc., Andover, MA, USA) as an intravenous bolus injection (1.34 mg d₃-leucine/kg body wt) followed by a continuous infusion (1.34 mg d₃-leucine/kg body weight/h) for 12 h. During infusion the subjects continued to fast but were allowed to drink water and remained ambulatory. Blood samples were drawn periodically for 10 h after bolus injection.

2.4. Isolation of lipoproteins and apolipoproteins

VLDL ($d < 1.006$ kg/l) and LDL ($1.019 < d < 1.063$ kg/l) were isolated from 5 ml plasma by sequential ultracentrifugation. Lp(a) was isolated from remaining plasma fractions by immunomagnetic separation with anti-apo(a) antibody 1A2-bound magnetic beads (MagnaBind™ carboxylated derivatized beads, Pierce, Rockford, IL, USA), as described previously [34]. Lp(a) was found to be pure, not degraded or fragmented and not contaminated with LDL. Likewise, purified VLDL and LDL did not contain Lp(a)/apo(a) [34].

2.5. Determination of isotope enrichment

Samples were prepared for GC–MS analysis as previously described [34]. Protein bands, corresponding to LDL-apoB as well as apo(a) and apoB from immunopurified Lp(a) were excised from the gel after preparative SDS polyacrylamide gel electrophoresis, dried overnight at 60 °C, hydrolyzed in 6 M HCl at 110 °C for 24 h and lyophilized. In case of subjects with two visible apo(a) bands, only the major apo(a) band was excised and analyzed.

2.6. Analysis of kinetic data using multicompartmental modeling

Tracer/tracee (T/T) ratios of apoB in VLDL and LDL, as well as apoB and apo(a) in Lp(a) were fitted to a multicompartmental model

(Fig. 1) using SAAMII (Washington University, Seattle, WA, USA), as previously reported [34,35]. Compartment 1 represents the plasma leucine pool and is used as a forcing function. Compartments 2 and 6 account for delay for apoB and apo(a), respectively. VLDL- and LDL-apoB pools consist of single compartments (3 and 4). Likewise, a single pool is allocated to Lp(a)-apoB (compartment 5) and Lp(a)-apo(a) (compartment 7). Values of $d(i,j)$ denote the distribution of transfer from delay compartment j to compartment i , whereas $k(i,j)$ represents the rate constant from compartment j to compartment i . The fractional catabolic rate (FCR) of VLDL-apoB-100 is equal to the sum of $k(0,3)$ and $k(4,3)$. FCR of LDL-apoB-100 is the sum of $k(0,4)$ and $k(5,4)$, assuming a conversion pathway from LDL to Lp(a) exists. Values of $k(0,5)$ and $k(0,7)$ represent FCR of apoB-100 and apo(a) in Lp(a), respectively. We assumed that steady-state conditions, under which FSR is equal to FCR were maintained throughout the study period. Retention time (RT) equals $1/\text{FCR}$. The production rate (PR) of a protein is equal to the product of $\text{FSR} \times (\text{plasma concentration of the protein}) \times (\text{plasma volume})$ divided by body weight [36]. Plasma volume was assumed to be 4% of body weight.

2.7. Statistical analysis

Data are presented as mean $\pm$ SD, except when indicated. For statistical analysis we used the nonparametric Mann–Whitney U test.

3. Results

Biometric data and lipoprotein profiles of the nine study subjects are shown in Table 1. All participants were white males with a mean age of 34 ± 10 years. Mean body mass index (BMI) at the beginning of the study was 24.5 ± 3.1 kg/m². Except subjects #6 and #9, study participants were normolipidemic, with mean plasma concentrations of 99 ± 49 mg/dl for triglycerides and 188 ± 47 mg/dl for cholesterol. Subjects #6 and #9 had elevated

plasma triglyceride and cholesterol concentrations of 216 mg/dl and 299 mg/dl, respectively. Lp(a) plasma concentrations were on average 42.3 ± 23.0 mg/dl and ranged from 16.6 to 92.1 mg/dl. Plasma lipid and apolipoprotein concentrations were determined in each plasma sample obtained during infusion and remained stable throughout, indicating steady-state conditions. Six subjects (#3–8) exhibited only one apo(a) protein isoform in plasma, whereas three (#1,2,9) displayed two different apo(a) isoforms. Apo(a) phenotyping in these three probands revealed a second larger isoform of markedly lower intensity (<5% of total staining), indicating a small and negligible contribution to the total Lp(a) plasma concentration. This second isoform was unlikely to significantly contribute to the apo(a) metabolism in these probands and was therefore not included in the analyses. In molar terms, mean Lp(a) plasma concentration of all probands was 123.9 ± 68.7 nM (Table 2). Apolipoprotein E phenotyping performed on delipidated plasma revealed six E3/3 carriers (#1,4,5,7–9), two heterozygous E4/3 carriers (#3,6) and one (#2) E2/2 carrier.

Mean FSR, RT and PR of apo(a) from Lp(a) were similar to those of apoB from Lp(a) but significantly different from the kinetic parameters of LDL-apoB (Table 2). The differences were particularly large between the PR values since this parameter takes into account plasma concentrations that are 15-fold higher for LDL than for Lp(a). Fig. 2 shows the average T/T curves for all subjects and demonstrates slower tracer enrichment of apo(a) and apoB from Lp(a) as compared to LDL-apoB. Fig. 3 shows an enlarged graph of T/T curves for apoB from Lp(a) and LDL for the very first 3 h investigated and demonstrates a contrasting rate of tracer enrichment for both proteins.

T/T data from Lp(a)-apo(a), Lp(a)-apoB, LDL-apoB and VLDL-apoB were analyzed based on the multicompartment model shown in Fig. 1 in order to investigate whether Lp(a) assembles from circulating LDL or from *de novo* produced “hepatic” LDL. Of leucine in Lp(a)-apoB $92.0 \pm 8.7\%$ originated from the hepatic apoB pool ($\text{FLUX}(5,2) = 1.3351$ nmol/h). The remaining 8% derived from plasma LDL-apoB ($\text{FLUX}(5,4) = 0.1047$ nmol/h) contributing 1.4398 nmol/h leucine ($\text{FLUX}(0,5)$). Apo(a) accounted for an average of 0.2396 nmol/h leucine ($\text{FLUX}(0,7)$). LDL-apoB stemmed from two sources, namely from VLDL-apoB ($\text{FLUX}(4,3) = 18.3568$ nmol/h leucine = 54%) and from *de novo* synthesis ($\text{FLUX}(4,2) = 15.6546$ nmol/h leucine = 46%).

Mean apo(a) FSR tended to be higher in subjects with HMW apo(a) isoforms (#1–4, 0.266 pools/day) than in those with LMW apo(a) carriers (#5–9, 0.221 pools/day, $p = 0.55$), without reaching significance. Finally, and consequently, the RT for apo(a) with HMW isoforms (3.90 days) tended to be shorter than for LMW isoforms (5.01 days, Table 2).

4. Discussion

The kinetic parameters obtained from this *in vivo* kinetic study of Lp(a) metabolism in healthy men allow two major conclusions: i) since the FSR of both protein components of Lp(a) was very similar and differed from those of LDL, an almost exclusive intracellular hepatic Lp(a) assembly can be assumed, ii) longer plasma RT for apo(a) from probands with LMW apo(a) isoforms as compared to those with HMW apo(a) isoforms aggravate the potential atherogenicity of the higher plasma concentrations.

Average FSR values for apo(a) in our study were comparable with those of previously published investigations using stable-isotope technology [22–24]. Analysis of kinetic data by multicompartmental modeling revealed separate pools of apoB from LDL and Lp(a). If Lp(a) were primarily synthesized from LDL, Lp(a)-apoB (product) T/T is expected to be lower than LDL-apoB (precursor) T/T. On the other hand, if Lp(a) was directly synthesized from the liver,

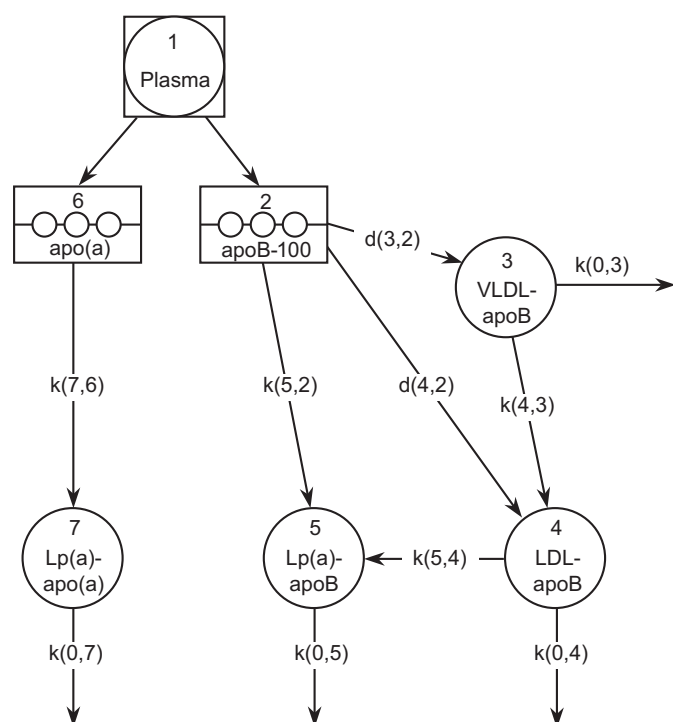


Fig. 1. Multicompartmental model for apoB-100 and apo(a) metabolism (for details, see Methods section).

Table 1

Biometric data, lipids, lipoproteins of participating subjects.

Subject	Age (yrs)	BMI (kg/m ²)	Triglycerides (mg/dl)	Cholesterol (mg/dl)	HDL-C (mg/dl)	LDL-C (mg/dl)	ApoB (mg/dl)	VLDL-apoB (mg/dl)	LDL-apoB (mg/dl)	Lp(a) (mg/dl)	ApoE isoform
1	43	25.4	57	170	42	117	98	13.50	78.4	36.2	3/3
2	23	22.5	100	150	37	93	150	3.00	143.0	16.6	2/2
3	36	30.6	124	196	45	126	119	1.17	111.0	34.0	4/3
4	42	24.6	50	154	52	92	99	0.40	95.3	19.5	3/3
5	21	22.0	90	187	56	113	94	1.48	82.5	60.7	3/3
6	50	27.6	216	204	36	125	125	5.20	111.6	49.0	4/3
7	27	22.3	86	141	34	90	67	4.00	48.0	92.1	3/3
8	28	21.0	79	190	62	116	48	0.26	41.7	37.2	3/3
9	32	24.2	91	299	41	240	194	1.60	186.2	35.2	3/3
Mean	34	24.5	99	188	45	124	110	3.40	99.7	42.3	
±SD	10	3.1	49	47	10	46	44	4.13	45.3	23.0	

Lp(a)-apoB T/T can be higher than LDL-apoB T/T, since LDL-apoB is the product of its precursors, VLDL-apoB. Fig. 3 shows indeed that Lp(a)-apoB T/T is higher than LDL-apoB T/T during the first 1.5 h, a finding consistent with an intracellular assembly concept. Indeed, analysis from the compartmental model demonstrated that most (on average 92%) Lp(a) apoB is directly synthesized from the liver (*k*(5,2) pathway) with only 8% remaining from the LDL (*k*(5,4)) pathway. This analysis, however, does not allow conclusions to be drawn on where (inside the hepatocyte, at its plasma membrane or, possibly, in the space of Dissé) this assembly takes place.

The *de novo* synthesis of LDL is an absolute prerequisite for the postulated (intra)cellular hepatic assembly of Lp(a). Such a “direct” LDL production has been questioned by some investigators who presume that it may instead be the consequence of a very fast lipolytic pathway [37]. However, kinetic studies of the apoB metabolism using stable-isotope technology fitted by multi-compartmental modeling support a significant “direct” LDL production by the liver [38,39]. A substantial amount of nascent LDL production was also detected in cultured primary human hepatocytes [40]. Lp(a) secretion was recently demonstrated in such cells, thus additionally supporting the assumption of “direct” LDL synthesis by human hepatocytes [41].

Most but not all *in vitro* studies that had so far investigated the mechanisms of Lp(a) assembly gave birth to the concept of extracellular assembly from newly secreted apo(a) with circulating plasma LDL, which may occur to some extent at the plasma membrane of the secreting hepatocyte [9]. Virtually all those

studies were conducted with transfected human hepatoma cell lines or baboon primary hepatocytes, both of which secrete a lipoprotein pattern significantly different from that of human plasma. In particular, baboon hepatocytes secrete most of their apoB as VLDL, which does not associate with apo(a) [42]. Hepatoma cells like HepG2, on the other hand, secrete a triglyceride-rich lipoprotein particle with an LDL density that does not exist at all in human plasma [43]. Taken together, the extracellular Lp(a) assembly proposed by several *in vitro* studies needs to be reviewed with caution, because those studies used cellular models that do not entirely reflect the physiological lipoprotein metabolism.

Previous kinetic studies also revealed contrasting results. The first radiotracer studies [20] found no radioactivity in circulating Lp(a) when radiolabeled VLDL was injected into humans and therefore suggested independent Lp(a) generation, which indirectly favors intracellular assembly. The substantially slower kinetics of Lp(a) (see results in Table 2) as compared with those of VLDL/LDL, however, do not allow this conclusion. While data obtained with radiotracer techniques are generally questionable because of the possibility of *ex vivo* manipulation (like radiolabeling of lipoproteins), a possible additional limitation of these studies could be the fact that in most cases the labeled Lp(a) investigated in individuals with apo(a) isoforms differed from the injected Lp(a) [6,7]. The first two studies using stable isotopes revealed similar FSRs for both Lp(a) protein components, again suggesting intracellular Lp(a) assembly. One study [22] investigated a very small group of subjects, whereas the other one [23] did not analyze the

Table 2

Kinetic data of study group.

Subject	Apo(a) isoforms	Lp(a) mol wt (kDa)	LP(a)-apo(a)				Lp(a)-apoB			LDL-apoB				VLDL-apoB			
			Lp(a) (nM)	FSR (pools/d)	RT (days)	PR (nmol/kg/d)	FSR (pools/d)	RT (days)	PR (nmol/kg/d)	LDL (nM)	FSR (pools/d)	RT (days)	PR (nmol/kg/d)	VLDL (nM)	FSR (pools/d)	RT (days)	PR (nmol/kg/d)
HMW																	
1	32/35	3580	101.1	0.210	4.76	0.849	0.240	4.17	0.973	1425	0.223	4.49	12.70	245.45	14.074	0.07	138.18
2	24/29	3450	48.1	0.229	4.37	0.441	0.202	4.95	0.389	2600	0.913	1.10	94.92	54.55	41.977	0.02	91.59
3	30	3546	95.9	0.309	3.27	1.186	0.493	2.03	1.892	2018	0.271	3.69	21.87	21.27	7.992	0.13	6.80
4	26	3480	56.0	0.314	3.19	0.704	0.568	1.76	1.273	1733	0.535	1.87	37.06	7.27	6.704	0.15	1.95
LMW																	
5	19	3370	180.1	0.267	3.82	1.888	0.183	5.47	1.316	1500	0.298	3.36	17.89	26.91	34.072	0.03	36.68
6	20	3385	144.8	0.188	5.32	1.090	0.205	4.88	1.185	2029	0.597	1.68	48.45	94.55	7.444	0.13	28.15
7	20	3385	272.1	0.214	4.67	2.332	0.203	4.93	2.205	758	0.308	3.25	9.34	72.73	9.804	0.10	28.52
8	13	3270	113.8	0.124	8.07	0.564	0.230	4.35	1.047	746	0.318	3.15	9.48	4.73	2.461	0.41	0.47
9	22/34	3420	102.9	0.316	3.17	1.301	0.367	2.73	1.511	3386	0.305	3.28	41.24	29.09	23.089	0.04	26.87
Mean			123.9	0.241	4.52	1.151	0.299	3.92	1.310	1799	0.418 ^a	2.87 ^a	32.55 ^{b,c}	61.84	16.402	0.12	39.91
±SD			68.7	0.066	1.54	0.623	0.143	1.38	0.527	843	0.223	1.09	27.46	75.11	13.690	0.12	45.92

FSR, fractional synthesis rate; RT, retention time; PR, production rate.

^a *p* = 0.04.^b *p* < 0.001 as compared to Lp(a)-apo(a).^c *p* < 0.001 as compared to Lp(a)-apoB.

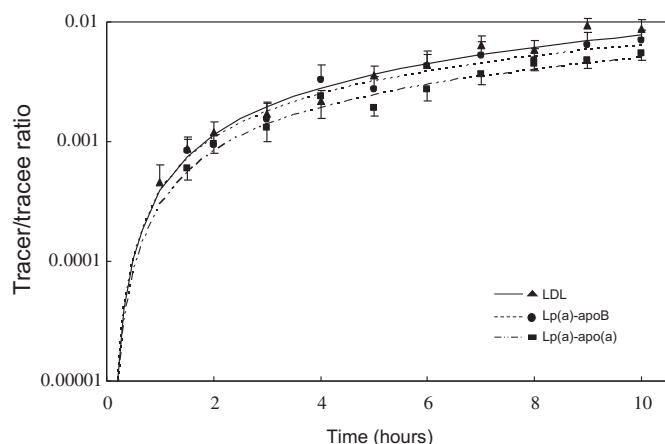


Fig. 2. Tracer/tracee (T/T) ratio curves for LDL-apoB-100 (triangles), Lp(a)-apoB-100 (circles), and Lp(a)-apo(a) (squares) in all subjects. Data are given as mean $\pm$ standard error and fitted by multicompartamental modeling using SAAMIII.

kinetics of LDL-apoB which is, however, needed to distinguish between the metabolic pathways of the two apoB-containing lipoproteins.

A further study by Demant et al. [24] using stable-isotope-labelled tracers and multicompartamental modeling came to the contrasting conclusion that at least half of the Lp(a) apoB moiety derives from preformed plasma lipoproteins. The reason for this discrepancy remains unclear and might stem from different study and/or Lp(a) purification protocols. The authors investigated only probands with LMW apo(a) isoforms and, in the first 10 h, found higher tracer enrichment rates for apo(a) and apoB of Lp(a) than for LDL-apoB. In addition, the enrichment curves of both Lp(a) components did not overlap, but steadily drifted apart. These observations contrast with our findings (Fig. 2). The expression of pool sizes (and consequently of production rates) in [mg] when comparing proteins with various molecular weights (apo(a) and apoB-100) is a limitation that we overcame in our study by converting mass into molar concentrations. An especially convincing argument against extracellular assembly is provided by the different starting times and the rates of tracer enrichment in apoB from Lp(a) and LDL which clearly demonstrate two separate pools for apoB as previously suggested (Fig. 3) [22]. The delayed appearance of tracer in LDL-apoB was obviously due to the finding

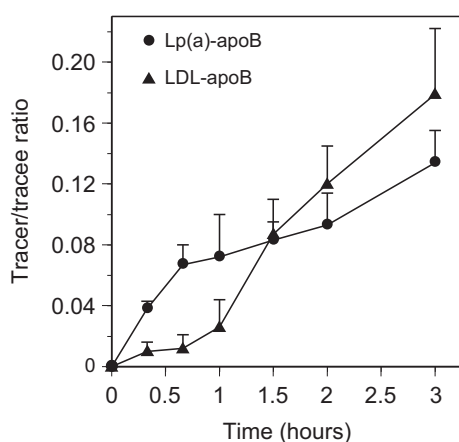


Fig. 3. Percentage tracer/tracee (T/T) ratio for Lp(a)-apoB (squares) and LDL-apoB (circles) from all subjects for the first 3 h of infusion and sample analysis.

that approximately half of LDL-apoB originated from lipolysis of triglyceride-rich lipoproteins.

Our results also disagree with conclusions drawn from a previously published kinetic study by Jenner et al. using a similar stable-isotope infusion protocol [25]. In contrast to our findings, Jenner et al. found significantly different FSR values for apo(a) and apoB of Lp(a), which is not compatible with an intracellular assembly concept. The reasons for these discrepancies might be the different purification protocols for Lp(a) from human plasma and the non-fasting state of probands in the study by Jenner et al.

Finally, the apo(a) isoform-dependent FSR accordingly tends to longer residence times for LMW apo(a) isoforms than for HMW apo(a) isoforms. This observation might add new momentum to the issue of the atherogenicity of Lp(a) with various apo(a) isoforms. Not only does Lp(a) with LMW apo(a) isoforms circulate on average at much higher plasma concentrations than does Lp(a) with HMW isoforms, but its RT in the bloodstream is also lengthened, which might substantially increase the atherogenicity of Lp(a). This is in line with the observation that LMW apo(a) phenotypes emerged as one of the strongest risk predictors for advanced stenotic atherosclerosis [44]. Moreover, LMW apo(a) isoforms were identified in a meta-analysis of prospective observational studies as being associated with a doubling of risk for CHD [45].

We are conscious of limitations in this kinetic study, which are itemized here. A difference in all kinetic studies exists in the dietary intake of the probands. The majority of stable-isotope studies investigating VLDL-IDL-LDL metabolism were performed in the fasting state. When compared with studies investigating probands in the fed state, kinetic parameters of LDL (but not of VLDL or IDL) differed substantially from those seen in fasting probands. The few Lp(a) kinetic studies conducted so far used protocols that differed with regard to the dietary state of their probands. While Morrisett et al. and Su et al. gave their probands, after overnight fast, light meals during the infusion period, Demant et al. investigated their patients in the fasting state during the first 10 h of infusion [22–24]. We are also aware of the relatively small number of participants in our study. Most kinetic studies, however, involve such small numbers of subjects due to the labor- and cost-intensive nature of these experiments.

In conclusion, this kinetic study using stable isotopes in nine male study subjects and fitted to a multicompartamental modeling indicates that Lp(a) is almost exclusively assembled intracellularly, which is well sustained by: i) two different apoB pools, ii) similar FCR, RT, and PR for Lp(a)-apo(a) and Lp(a)-apoB, suggesting that both proteins have similar, if not identical, metabolic pathways, and iii) different FCR, RT and PR of Lp(a)-(Lp(a)-apo(a) and Lp(a)-apoB, respectively) as compared to LDL-apoB. Future studies of human primary hepatocytes should permit the precise location of assembly to be determined. Finally, carriers of LMW apo(a) isoforms reveal not only higher plasma Lp(a) concentrations but also longer Lp(a) RT in plasma, which might substantially contribute to the higher atherogenicity of LMW apo(a) as compared to that of HMW apo(a).

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